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उत्पाद मानुयल
TEXTILES —HIGH DENSITY POLYETHYLENE (HDPE)
/POLYPROPYLENE (PP) WOVEN SACKS FOR
PACKAGING OF 50 KG CEMENT
IS 11652:2017 के अनुसार

PRODUCT MANUAL For
TEXTILES — HIGH DENSITY POLYETHYLENE (HDPE)
/POLYPROPYLENE (PP) WOVEN SACKS FOR
PACKAGING OF 50 KG CEMENT
ACCORDING TO IS 11652:2017

भारतीय मानक ब्यूरो (अनुरूपता मूल्यांकन) विनियम की स्कीम-I के तहत यह उत्पाद मानुयल प्रमाणीकरण के प्रचलन में रीति और पारदर्शिता के सुसंगत सुनिश्चित करने के लिए सभी क्षेत्रीय/शाखा कार्यालयों एवं लाइसेन्स धारियों द्वारा संदर्भ सामग्री के रूप में उपयोग किया जाएगा। बीआईएस लाइसेन्स/प्रमाण पत्र प्राप्त करने के इच्छुक भावी आवेदकों द्वारा भी इस दस्तावेज़ का उपयोग किया जा सकता है।

This Product Manual shall be used as reference material by all Regional/Branch Offices & licensees to ensure coherence of practice and transparency in operation of certification under Scheme-I of Bureau of Indian Standards (Conformity Assessment) Regulations, 2008 for various products. The document may also be used by prospective applicants desirous of obtaining BIS certification licence/certificate.

1.	उत्पाद Product	:	IS 11652:2017
	शीर्षक Title	:	TEXTILES — HIGH DENSITY POLYETHYLENE (HDPE) /POLYPROPYLENE (PP) WOVEN SACKS FOR PACKAGING OF 50 KG CEMENT -SPECIFICATION
	संशोधन संख्या No. of amendments	:	01
2.	नमूनाकरण दिशा निर्देश Sampling Guidelines		
a)	कच्चा माल Raw material	:	The high density polyethylene (HDPE) or polypropylene(PP) used for manufacture of tape shall be virgin and confirm to the requirements specified in IS

		10146 or IS 10910, respectively excluding overall migration. If the firm is directly purchasing HDPE/PP Fabric Tapes, these shall confirm to Clause 4.2 of IS 11652, and IS 6192 / IS 11197 respectively.
b)	समूहिकरण दिशा निर्देश Grouping Guidelines	: Separate sample shall be tested for HDPE sacks and PP sacks. If sacks with lamination / Gusseted / UV Stabilization are tested, sacks without lamination / without UV Stabilization /Non Gusseted respectively may also be covered in the scope. Scope shall be restricted based on manufacturing and testing capability and samples of different varieties shall be drawn in rotation during surveillance inspections.
c)	नमूने का परिमाण Sample Size	: 10 sacks + approx 15m X 6 reels HDPE/PP tape + approx 20 m stitching thread + approx 1m X 1m un-laminated fabric + approx 1 m X 1m laminated fabric.
3.	परीक्षण उपकरणों की सूची List of Test Equipment	: Please refer Annex –A
4.	निरीक्षण व परीक्षण स्कीम Scheme of Inspection and Testing	: Please refer Annex - B
5.	एक दिन में संभावित परीक्षण Possible tests in a day	: When appropriately conditioned samples are available, the following tests can be carried out in a day: (i) Linear density of tape (Cl. 4.2 and IS 6192/IS11197) (ii) Tenacity of tape (Cl. 4.2 and IS 6192/IS11197) (iii) Elongation of tape (Cl. 4.2 and IS 6192/IS11197) (iv) Thickness of tape (Cl. 4.2 and IS 6192/IS11197) (v) Width of tape (Cl 4.2 and IS 6192/IS 11197) (vi) Tests for Bottom Seam(Cl.4.3.1) (vii) Lamination Mass (Cl.4.4) (viii) Lamination Overhang (Cl.4.4) (ix) Dimensions [Cl. 5.2 and Table 1(i)] (x) Ends per dm [Cl. 5.2 and Table 1(ii)] (xi) Picks per dm [Cl. 5.2 and Table 1(iii)] (xii) Mass of sack, gusseted type [Cl. 5.2 and Table 1(iv)] (xiii) Breaking strength of bottom seam [Cl. 5.2 and Table 1(vi)] (xiv) Drop Impact Strength [Cl. 5.2 and Table 1(viii)]

6.	लाइसेन्स का कार्यक्षेत्र/Scope of the Licence:	
	IS 11652:2017 के अनुसार मानक मुहर का उपयोग करने के लिए लाइसेन्स निम्नलिखित कार्यक्षेत्र के लिए प्रदान किया जाता है Licence is granted to use Standard Mark as per IS 11652:2017 with the following scope:	
	Name of the product	TEXTILES — HIGH DENSITY POLYETHYLENE (HDPE) /POLYPROPYLENE (PP) WOVEN SACKS FOR PACKAGING OF 50 KG CEMENT
	Material	High Density Polyethylene (HDPE) / Polypropylene (PP)
	Type	With / without Lamination, with / without UV Stabilization, Gusseted / Non Gusseted

ANNEX-A
LIST OF TEST EQUIPMENTS

Major test equipment required to test the requirements of the Indian Standard.

Sr. No.	Tests used in with Clause Reference	Test Equipment
1	Tapes – Width and Thickness (Cl. 4.2 and IS 6192/IS 11197)	Steel Rule / Vernier 0.1mm LC, Spring Loaded dial micrometer of 0.001mm LC & 140kN/m ² pressure
2	Tapes – Linear Density (Cl. 4.2 and IS 6192/IS 11197)	Weighing Balance 0.1mg accuracy, Drying Oven, Wrap Reel
3	Tapes – Tenacity (Cl. 4.2 and IS 6192/IS 11197)	Tensile testing machine with constant rate of extension, suitable clamps and facility to measure elongation
4	Tapes – Elongation at point of rupture (Cl. 4.2 and IS 6192/IS 11197)	Tensile testing machine with constant rate of extension, suitable clamps and facility to measure elongation
5	Tapes – Heat Shrinkage (HDPE) (Cl. 4.2 and IS 6192)	Heat shrinkage test apparatus with water bath, specimen holder and lead in wire
6	Tapes – Colour Fastness for pigmented tapes (Cl. 4.2 and IS 6192/IS 11197)	Xenon Arc lamp test apparatus, Geometric grey scale
7	Tapes – Dimensional stability (PP) (Cl. 4.2 and IS 11197)	Air circulating Oven
8	Unlaminated Fabric Mass (Cl. 4.2)	Weighing Balance 5mg accuracy, Steel Scale, Flat smooth table
9	Seam Stitching (Cl. 4.4)	Steel Scale
10	Material used for stitching (If different from HDPE/PP Tape)- Breaking Load (Cl.4.4)	Tensile testing machine
11	Lamination - Mass, Overhanging (Cl. 4.5)	Weighing Balance, Steel scale, Flat smooth table
13	Capacity (Cl.4.6)	Weighing Scale, Filling Material
14	Mass of bale (Cl. 5.1)	Weighing Scale
15	Dimensions [Cl. 5.2 and Table 1 i)]	Scale, Flat smooth table
16	Ends per dm [Cl. 5.2 and Table 1 ii)]	Pick glass
17	Picks per dm [Cl. 5.2 and Table 1 iii)]	Pick glass
18	Mass of sack, [Cl. 5.2 and Table 1 iv]	Weighing Balance,
19	Average breaking strength of fabric [Cl. 5.3 and Table 1 v)]	Tensile testing machine CRE with facility to record/indicate force and separation
20	Breaking strength of top & bottom seam [Cl. 5.3 and Table 1 vi)]	Tensile testing machine with suitable clamps, stop watch, scale

21	Elongation at break of fabric [Cl. 5.2 and Table 1 vii)]	Tensile testing machine CRE with facility to record/indicate force and separation
22	Drop impact strength [Cl. 5.4 and Table 1 viii)]	Drop Impact test apparatus, filling material
23	Ash content [Cl. 5.5 and Table 1 ix)]	Weighing Balance 1mg accuracy, Silica crucible, Bunsen Burner, Silica Triangle and Tripod, Muffle furnace, Desicator, Gloves, Crucible holder
24	UV resistance (Cl. 5.6)	Test Chamber with Fluorescent UV- lamps Type B with irradiance level 0.63W/m ² with facility for heating and condensation, Tensile testing machine CRE type
25	Conditioning of sample (Cl. 7)	Conditioning Chamber to maintain 27±2°C, 65±2% humidity

The above list is indicative only and may not be treated as exhaustive

ANNEX – B**SCHEME OF INSPECTION AND TESTING**

- 1. LABORATORY** - A laboratory shall be maintained which shall be suitably equipped (as per the requirement given in column 2 of Table 1) and staffed, where different tests given in the specification shall be carried out in accordance with the methods given in the specification.

- 1.1** The manufacturer shall prepare a calibration plan for the test equipment.

- 2. TEST RECORDS** – The manufacturer shall maintain test records for the tests carried out to establish conformity.

- 3. PACKING AND MARKING** – The Standard Mark as given in the Schedule of the license shall be printed using suitable inks by flexography or gravure printing or stenciling on each sack and bale of HDPE /PP woven sacks; as the case may be, provided always that sacks in each bale to which this mark is thus applied conforms to every requirement of the specification.

Note: The words “for HDPE woven sack only” or “ for PP woven sack only” as applicable shall be printed beneath the standard mark conspicuously on each sack.

- 3.1** Marking and packing shall be done as per the provisions of the Indian Standard. In addition, the following information shall be given on each bale or the label attached to it:

a) Control Unit (or Batch) Number.

b) BIS website details i.e.–“For details of BIS certification please visit www.bis.gov.in”

- 4. CONTROL UNIT** – For the purpose of this scheme, All the HDPE/PP woven sacks of the same construction produced under similar conditions in a day shall constitute a control unit.

- 5. LEVELS OF CONTROL** - The tests as indicated in column 1 of Table 1 and the levels of control in column 3 of Table 1, shall be carried out on the whole production of the factory which is covered by this plan and appropriate records maintained in accordance with paragraph 2 above.

- 5.1** All the production which conforms to the Indian Standard and covered by the licence should be marked with Standard Mark.

- 5.2 SAMPLING FOR FLOOR INSPECTION**

Sacks shall be selected during production from each a control unit and shall be tested as under::

Sl. No.	Characteristics	Sample Size	Frequency	Total No. of Samples
1	Dimensions	5 sacks	Every 4 hours	30 sacks

2	Ends & Picks per dm	5 sacks	Every 4 hours	30 sacks
3	Breaking strength of Fabric, top & bottom seam strength, and elongation at break	1 sack	One sample when control unit starts and then at equal intervals of 2 hrs	05 sacks
4	Sack, Seam, Lamination(If required by the buyer), Capacity and Valve for filling of sacks	5 sacks	Every 4 hours	30 sacks
5	Mass of the sack	5 sacks	Every 4 hours	30 sacks

Notes:

- Each bundle, which is an intermediate packaging, normally contains 50 sacks. However, the number of sacks in each bundle may vary as per the customer's requirement.
- In case of any non-conformity, the production to be stopped, Root cause analysis to be carried out and corrective actions to be taken before resuming production.
- In the event of failure of the sample in one or more of the requirements, the entire material in the control unit may be either reprocessed for rectifying the defect or rejected. Such reprocessed material when tested again shall conform to all the requirements of the specification. -
- REJECTIONS**—Disposal of non-conforming product shall be done in such a way so as to ensure that there is no violation of provisions of BIS Act,2016.

TABLE 1
LEVELS OF CONTROL

(1)				(2)	(3)		(4)
TEST DETAILS					LEVELS OF CONTROL		Remarks
Cl. No.	Requirements	Test Method		Test equipment requirement R: required (or) S: Sub-contracting permitted	No.of Samples	Frequency	
		Clause	Reference				
4.1	Raw Material (HDPE/PP granules)	4.1	IS 11652	S	01	Each consignment	One sample from each consignment shall be tested for ascertaining conformity to IS 10146 or IS 10910, excluding over all migration, unless the same is accompanied by test certificate from supplier.
4.2	Fabric	4.2	-do-	R	05	Every 4 hours production (If tapes are manufactured in house)	Samples shall be drawn and tested at regular intervals to ascertain their conformity to the requirements of Cl. 4.2
					05	Each consignment (if tapes are procured from outside)	No testing is required if consignment is accompanied with test certificate of supplier indicating conformity to the specified requirements or is ISI marked.
4.3	Sacks	4.3	-do-	R	30 sacks	5 sacks Every 4 hours	PI see 5.2 of SIT
4.4	Seam	4.4	-do-	R	30 sacks	5 sacks Every 4 hours	-do-
4.4.	Material used for Stitching	4.4	-do-	S	01	Each consignment	No testing may be required if consignment is accompanied by test certificate of supplier indicating conformity to Cl 4.4
4.5	Lamination	4.5	-do-	R	30	5 sacks Every 4 hours	Only if required by buyer (PI see 5.2 of SIT)

4.6	Capacity	4.6	-do-	R	30	5 sacks Every 4 hours	PI see 5.2 of SIT
4.7	Valve for filling of sacks	4.7	-do-	R	30	5 sacks Every 4 hours	PI see 5.2 of SIT
5	REQUIREMENTS						
5.1	Mass of Bale		IS 1964	R	Each bale		
5.2, Table 1	Requirements of HDPE/PP Woven Sacks for Packaging Cement						
(i)	Dimensions	Annex B	IS 11652	R	30 sacks	Each control unit	PI see 5.2 of SIT and Note 1 under Table 1 of IS 11652
(ii)	Ends per dm	-do-	-do-	R	30 sacks	Each control unit	PI see 5.2 of SIT
(iii)	Picks per dm	-do-	-do-	R	30 sacks	Each control unit	PI see 5.2 of SIT
(iv)	Mass of Sack		IS 1964	R	30 sacks	Each control unit	The mass of sack with dimensions other than those specified shall be calculated by the method given in Annex E . Also PI see 5.2 of SIT
(v)	Average Breaking Strength of Fabric		IS 1969 (Part 1)	R	5 sacks	Each control unit	PI see 5.2 of SIT
(vi)	Breaking strength of top and bottom seam		IS 9030	R	5 sacks	Each control unit	PI see 5.2 of SIT
(vii)	Elongation at break of fabric		IS 1969 (Part 1)	R	5 sacks	Each control unit	PI see 5.2 of SIT
(viii)	Drop Impact Strength	Annex C	IS 11652	R	5 sacks	Each control unit	
(ix)	Ash Content	Annex D	-do-	R	2 sacks	Each control unit	
5.6	UV Resistance	Annex F	-do-	S	For UV Resistance test which is to be done if agreed to between buyer and seller as per the IS, the agreement between buyer and seller for this test may specify the frequency of the test.		

Note-1: Whether test equipment is required or sub-contracting is permitted in column 2 shall be decided by the Bureau and shall be mandatory. Sub-contracting is permitted to a laboratory recognized by the Bureau or Government laboratories empaneled by the Bureau

Note-2: Levels of control given in column 3 are only recommendatory in nature. The manufacturer may define the control unit/batch/lot and submit his own levels of control in column 3 with proper justification for approval by BO Head.